

Validation and Testing Plan

Quality Assurance Strategy — StockWise Project

Testing Levels & Methods

Testing Type	What is Verified	Tool / Method
Unit Testing	Service layer methods tested in isolation	JUnit 5 + Mockito
Integration Testing	API endpoints + frontend–backend flow	Postman + @SpringBootTest
Manual Testing	UI behavior, edge cases, role-based access	Browser + Jira bug tracking
Security Testing	SQL injection, broken auth, access control	Postman (manual probing)
Performance Testing	API response time under expected load	Postman Runner

Testing Timeline

- Unit tests written alongside development (1 test per service method).
- Integration testing, manual UI testing, bug fixing.
- Regression tests, final report, all results pushed to GitHub.

Responsibilities

- Can Tasar (QA) — test plan, unit tests, integration tests, bug tracking.
- Tuba Eraslan (Backend) — writing service-layer unit tests, fixing backend bugs.
- Ceren Şengül (Frontend) — manual UI testing, cross-browser verification.

Success Criteria

Success Criterion	How It's Verified
Unit test coverage ≥ 80%	JUnit + Jacoco report
API response time < 500ms	Postman timing
JWT auth blocks unauthorized users	Role-based Postman tests
Stock alert fires within 5 seconds	Manual + integration test
Zero critical (P1) bugs at delivery	Jira bug tracker
All documentation on GitHub	Repository review